

EXP. NO: 7

Data Visualization Using Tableau

AIM:

To become familiar with the Tableau workspace and build visual analyses from a CSV data source. The experiment covers creating bar, line, and pie charts, defining a calculated field for profit ratio, and combining the worksheets into an interactive dashboard and a data story.

PROCEDURE:

1. **Start Tableau and Explore the Workspace**
 - i. Open Tableau Desktop or Tableau Public.
 - ii. Review the Start Page, Data pane, Worksheet, Dashboard, and Story tabs.
 - iii. Locate the Columns and Rows shelves, Marks card, Filters shelf, and Show Me panel.
2. **Connect to the Data Source**
 - i. Choose Connect → Text File.
 - ii. Browse to Tableau_SalesData.csv and open it.
 - iii. Check that the fields Date, Region, Category, Sales, Profit, Quantity, and Target are listed in the Data pane.
3. **Build the Basic Charts**
 - i. Bar chart: drag Sales to Columns and Category to Rows, then sort in descending order and show mark labels.
 - ii. Line chart: drag Date (as Month) to Columns and Sales to Rows, and add markers to the line.
 - iii. Pie chart: set the Marks type to Pie, drag Region to Color and Sales to Angle.
4. **Create the Calculated Field**
 - i. Select Analysis → Create Calculated Field.
 - ii. Name the field Profit Ratio % and enter the formula: $\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales}]) * 100$
 - iii. Build a bar chart of Profit Ratio % by Category.
5. **Design the Interactive Dashboard**
 - i. Create a new dashboard and name it Regional Sales Performance Review.
 - ii. Drag in the four worksheets: Sales by Category, Monthly Sales Trend, Sales by Region, and Profit Ratio by Category.
 - iii. Add a Region filter (as a dropdown) and apply it to all worksheets that use this data source.
 - iv. Resize and position the sheets in a 2×2 grid.
6. **Create the Story**
 - i. Create a new Story.
 - ii. Add the dashboard as the first story point and add a caption to it.
 - iii. Set the story title to Overall Sales Analysis.

OUTPUT:

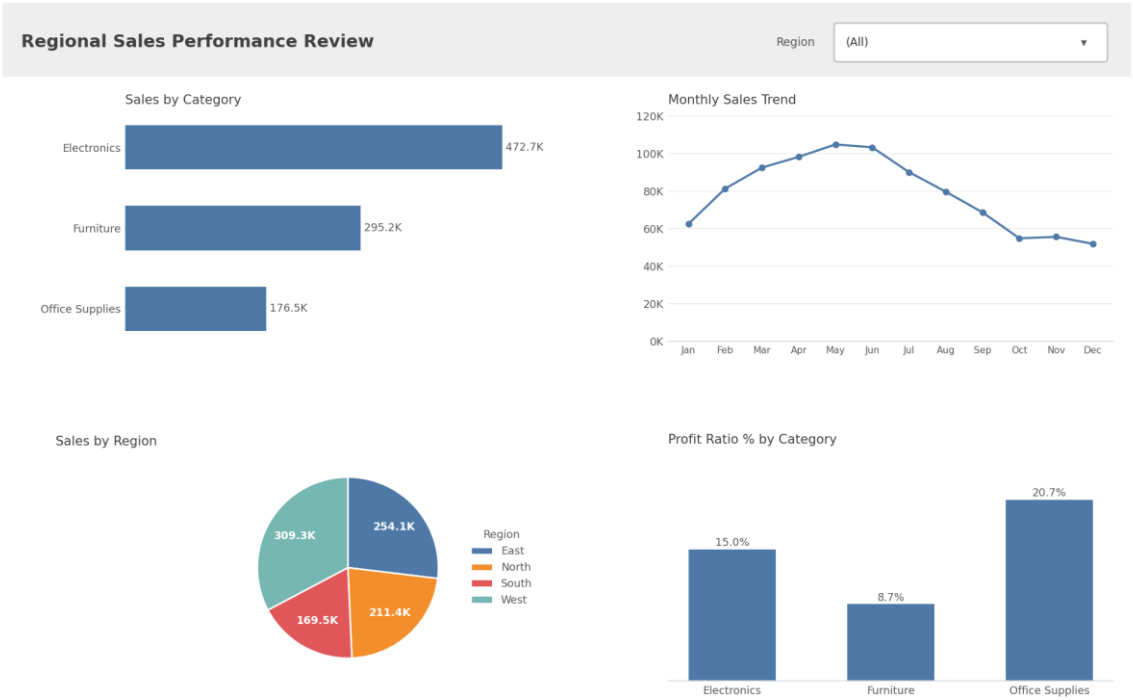


Fig 1: Dashboard – Region: All

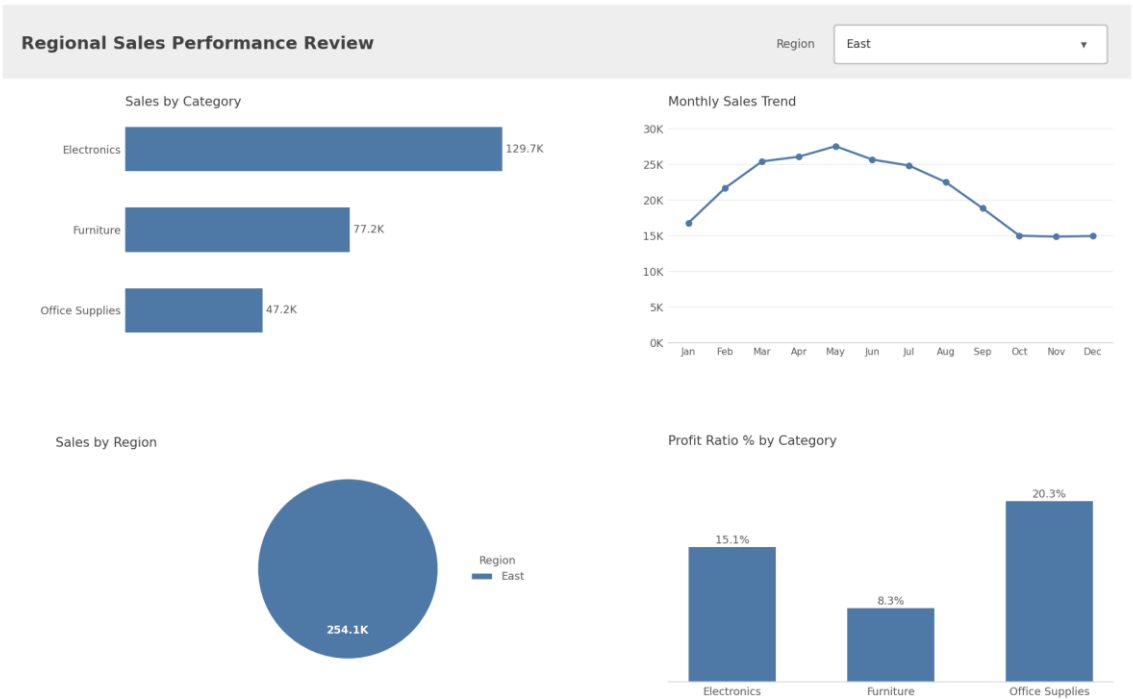


Fig 2: Dashboard – Region: East

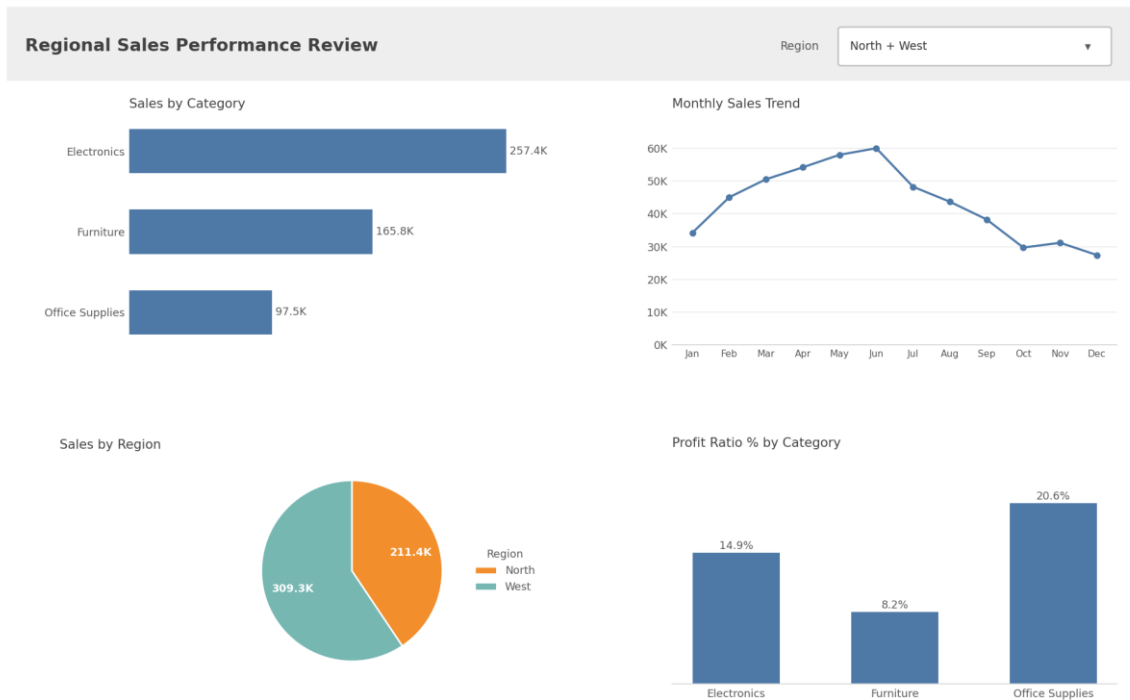


Fig 3: Dashboard – Region: North + West

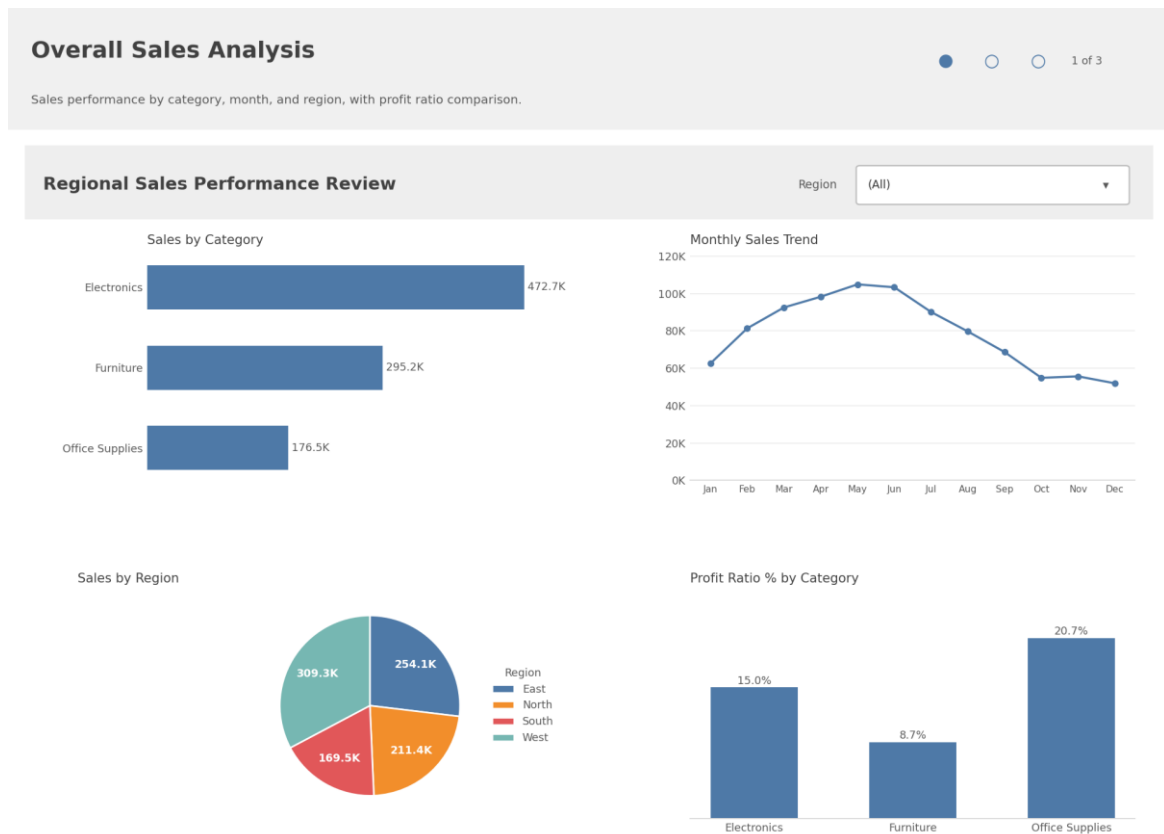


Fig 4: Story – Overall Sales Analysis

RESULT:

Thus, the sales data was loaded into Tableau and analysed successfully. Bar, line, and pie charts were built along with a Profit Ratio % calculated field, and the results were presented through an interactive dashboard and a story.